

# RAJAT SAVDEKAR

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## EDUCATION

University of Southern California, Los Angeles

Master of Science, Computer Science

Aug 2024 – May 2026

GPA: 3.63

University of Mumbai, Mumbai

Bachelor of Engineering, Computer Engineering

Aug 2019 – May 2023

GPA: 3.7

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C++, Swift, SQL, HTML/CSS

**Web & Frameworks:** React, Next.js, Node.js, Express, Flask, FastAPI, REST APIs, WebSockets

**Data & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, LevelDB, Yjs (CRDTs)

**Cloud & Infrastructure:** AWS, Docker, Docker Compose, Terraform, Linux, CI/CD, Git

**AI & Systems:** LLMs, RAG, OpenAI API, Model Quantization (bitsandbytes), PyTorch, Whisper

## EXPERIENCE

### Software Developer

Easley-Dunn Productions, Inc.

Aug 2026 – Present

Torrance, CA

- Reduced AI-narration inference cost by **60%** across **150+ surgical replays** by 4-bit quantizing a 7B medical LLM, batching speech synthesis, and caching model plus translation outputs.
- Cut replay upload failures by replacing **100 MB** in-memory SQLite blobs with streamed object storage and a file API that matches the Quest client.
- Increased Quest training scenes frames to **72 Hz** and cut drill time **4x** by stripping 255 extra mesh-deform scripts, then added CI/CD that deploys Quest builds, which caught **8** regressions before merge.

### Teaching Assistant

USC Summer Programs

May 2025 – Jul 2025

Los Angeles, CA

- Led **international teams** from 100 students through Python, Github, Linux and basic web development (HTML/CSS) and Arduino C / C++ , guiding them to build and present project websites and earn A+ grade.

### Software Engineer

Jalgaon Fruit Sales Cooperative Ltd – Agritech Supply Chain Platform

May 2023 – May 2024

Jalgaon, India

- Decomposed a legacy PHP, SQL monolith by building Python Flask microservices backed by PostgreSQL, orchestrating automated cron jobs to synchronize state and cutting query latency by **40%** across **1,000+ daily truck transactions**.
- Doubled peak harvest write throughput (**2x**) and scaled revenue from **\$100K to \$200K** by architecting an isolated write path that eliminated 10–15 minute freezes from MyISAM table locking.
- Delivered a web procurement platform with RESTful APIs and live analytics dashboards for **150+ daily clerks** across 5 warehouses, reducing deploy cycles by **30%** by containerizing services with Docker, pytest validation, and CI/CD.

## SELECTED PROJECTS & OPEN SOURCE

### MockPad – Real-Time Collaborative Engine & Multimodal AI Replay

[\[Live\]](#) [\[GitHub\]](#)

- Architected a real-time collaborative editor on Yjs CRDTs over custom WebSockets, sustaining **sub-100ms sync latency** for **30+ active users** deployed on a single \$6/month DigitalOcean droplet.
- Cut server peak memory consumption by **80%** with **zero session data loss** by implementing an LRU disk-eviction strategy to LevelDB for inactive collaborative rooms.

### CloudBridge – AI Cloud Architecture Advisor & Topology Engine

[\[GitHub\]](#)

- Engineered an LLM-driven synthesis engine in FastAPI mapping natural language constraints into validated AWS topologies with IAM guardrails and automated Terraform export against the **Well-Architected Framework**.
- Built a reactive DAG topology generator in React Flow and TypeScript, delivering **sub-100ms** real-time parameter mutations, cost projections, and multi-service dependency recalculations.

### Caliberate – LLM Code-Eval Harness

[\[GitHub\]](#)

- Architected a distributed LLM evaluation harness using network-isolated Docker workers, ensuring zero-loss task execution via a PostgreSQL SKIP LOCKED queue and lease-based crash recovery.
- Integrated an LLM-as-judge verifier into CI/CD, calibrating it against **180** hand-labeled samples to automatically block merges if prompt or model updates caused accuracy regressions.

## HACKATHONS & PUBLICATIONS

**Aura – visionOS Accessibility Assistant.** Aug 2025. 2nd place of 24-hour hackathon; built with Swift, SwiftUI, and RealityKit, delivering fully on-device live captions and environmental sound awareness. [\[GitHub\]](#)

**Research Publications.** *Texture Feature Analysis Using GLCM Matrix (10 citations)*; medical-transcript NLP; LSB image steganography. [\[GitHub\]](#)